Title

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June 1, 2026

Abstract

This is the abstract. It should be no more than 250 words and should not contain any references or displayed equations. The abstract should be a single paragraph and should not be subdivided into sections. The abstract should provide a concise summary of the paper, including the motivation, methods, results, and conclusions.

Keywords: 3 to 6 keywords, that do not appear in the title

1 Introduction

Reporting the sequence of steps and decisions taken in a data analysis is a crucial part of presenting evidence in publications for new scientific discoveries. Modern data analyses have grown in complexity, with data coming from multiple sources and are potential spatial, temporal, and multivariate nature. To ensure the reproducibility of reported results, good practices are advocated to make code scripts publicly available or adopt literate programming techniques. To this date, reproducibility checks by the journals have been a standard screening process in the publication pipeline in statistics.

One implicit assumption the current practices operates under is that an analysis becomes reproducible, hence understandable, once all the reproducible materials are made available. Specifically, understandable implies that an independent researcher should be able to, without much difficulty, track the steps that have been done on the data and understand why each step was taken. In practice, however, reconstructing an analysis is rarely straightfoward. Instead, it is often a frustrating cycle of trial and error, hitting failure points with little feedback on when things go wrong. Part of the reason is that the two media (manuscripts and code scripts) serve different communication goals. A manuscript is a curated narrative designed to present new scientific discoveries, rather than an algorithmic log of the data analysis workfllow. Code scripts, on the other hand, show all the computational details but don't inherently carry the underlying scientific rationale behind each step. Additionally, code scripts reflect author's choice of pragramming language, personal coding style, and customized abstractions built to streamline repetitive analytical tasks. These all create obstables for readers attempting to parse the code and hinder their ability to confidently modify or extend the analysis.

Hence, a gap still exists between an analysis being machine replicable and reader understandable, requiring researchers to spend a non-trivial amount of time piecing together information from the manuscript and code scripts. With recent advances in large language models (LLMs), we have the opportunity to rethink how we can present data analysis in a way that is more structured and to facilitate comprehension and exploration of alternatives. LLMs have shown remarkable capabilities in understanding human language and in generating code from natural language descriptions. This opens up new possibilities to bridge the gap from reproducibility to understandability.

In this paper, we present a schema designed to describe data analysis in a structured way that captures the key decisions and their justifications, as well as the inputs and outputs of each step. We also demonstrate how this schema can be used to generate code scripts from textual descriptions of analyses, and to explore alternative analytical paths given the power of LLMs for code generation. We apply our schema to a case study of the many-analyst football study, where we extract the analytical decisions from the final reports of each team, generate the corresponding code scripts, and validate the results against the published results in the original paper. We also demonstrate how to specify and explore alternatives for a specific step in the analysis using our schema.

The paper is organized as follows. In Section 2, we review the relevant literature on communication of data analysis and reproducibility, as well as agentic data science. In Section 3, we present our methodology, including the schema for describing data analysis and the process for generating and validating code from the schema by LLMs. In Section 4, we apply our methodology to the many-analyst football study and present our results. Finally, in Section 5, we discuss the implications of our work and potential future directions.

2 Literature review

2.1 Communication of data analysis and reproducibility

2.2 Agentic data science

3 Methodology

3.1 A schema for describing data analysis

We represent a data analysis, $\mathcal{S} = \{A_1, A_2, \cdots, A_n\}$, as a sequence of n analytical step, A_i, applied to a dataset $\mathcal{D}$, where each step A_i is defined as a tuple:

$$A_i = \langle f_i, \pi_i, r_i, \mathcal{I}_i, \mathcal{O}_i, \omega_i \rangle. \tag{1}$$

The components of A_i are partitioned into a required triplet (f_i, π_i, r_i) and a set of optional components $(\mathcal{I}_i, \mathcal{O}_i, \omega_i)$. The first component of the triplet, $f_i \in \mathcal{F}$, defines the **objective** of a data analysis step that invites multiple justifiable approaches. The second component, $\pi_i \in \Pi$, represents the data analysis **decision** made to achieve the objective f_i, which can be read from both the manuscript and from the code script. The term r_i is the **justification** provided for choosing the decision π_i. This field is usually provided in the manuscript but not the code script to show the scientific intent of the step. As an example, consider the three steps involved in calculating the Human Development Index (HDI):

1) Scale all four variables (life expectancy, expected years of schooling, mean years of schooling, and GNI per capita) to a [0, 1] range using their corresponding upper and lower limits.

2) Combine the two education-related variables (expected years of schooling and mean years of schooling) into a single education variable.
3) Combine the three core dimensions (health, education, and income) into a single index using a geometric mean.

Structuring this information into a schema is involves defining the objective, decision, and justification for each step. While the decision (what is to be done) and the justification (why it is done) are generally clear, the objective requires identifying what the step is trying to achieve. In other words, it represents the broader goal that could potentially be met through alternative methods (formally introduced in Section 3.2). For instance, in step 3, the objective is to combine the three dimensions. While a simple arithmetic mean could be an alternative solution for this goal, the geometric mean is specifically chosen because it penalizes uneven development among the three dimensions—this reasoning forms the justification. This process results in the basic schema shown in Listing 1:

On the optional components of a data analysis step A_i, the operational flow of the analysis is governed by the input and output spaces, $\mathcal{I}_i$ and $\mathcal{O}_i$, once a dataset is provided. While the output of a step can be a pseudo object that not currently exist, the inputs for any given step must be within the existing data scope, comprising variables in the original dataset and all outputs from previous steps. Formally, we require:

$$\mathcal{I}_i \subseteq \left(\mathcal{D} \cup \bigcup_{j<i} \mathcal{O}_j \right), \quad \forall i \in \{2, \dots, n\}.$$

A schema can also be automatically extracted by Large Language Models (LLMs) from text. If a dataset is provided, the inputs and outputs will also automatically inferred during the extraction. We also introduce a categorical confidence score, ω_i (low, medium,

Listing 1 An example schema that specifies the steps in calculating Human Development Index (HDI).

```
meta:
  type: schema
nodes:
- tag: step-scaling
  objective: how to handle variables in different scales?
  decision: apply min-max scaling to each variable
  rationale: to convert the variables to the same scale for future construction of a si
- tag: step-education
  objective: how to combine the two education-related variables?
  decision: take the average
  rationale: the arithmetic mean allows perfect substitutability between the two variab
- tag: step-combine
  objective: how to combine the three dimensions into a single index?
  decision: use the geometric mean
  rationale: the geometric mean penalizes uneven development among the three core dimens
```

high) that can be useful in case when the source text is ambiguous to guide the LLM to ask for clarification to consolidate the schema extraction.

3.2 Describing alternatives

While one analysis represents a realized series of choices, alternatives can often be taken for to address the same objective, each with its own set of assumptions and implications. For each step i in the sequence, to fulfill each fork f_i, there could be k_i decision choices to choose from: $\{A_i^1, A_i^2, \ldots, A_i^{k_i}\}$. We can define a selection vector $r = (r_1, r_2, \ldots, r_n)$,

where $r_i \in \{1, \ldots, k_i\}$, to describe an analysis, comprised of specific series of decisions: $\mathcal{S}^r = \{A_1^{r_1}, A_2^{r_2}, \ldots, A_n^{r_n}\}$. A multiverse, used to describe the entire decision spae, is defined as the Cartesian product of these sets: $\mathcal{M} = \{\mathcal{S}^r \mid r_i \in 1, \ldots, k_i, \forall i\}$. The cardinality of this space, $|\mathcal{M}| = \prod_{i=1}^{n} k_i$, quantifies the total number of unique, fully-specified analysis.

Returning to the HDI example, in step 3, instead of a geometric mean, an arithmetic mean to combine the three dimensions into a single index. It may not be the best decisions, but still a plausible one to achieve the objective of this step. Listing 2 demonstrates how this single alternative is structured in an alternative YAML file. Additional alternatives can be specified by adding items to the `alternatives` list.

Listing 2 An example YAML file for specifying alternative for Step 3 in calculating Human Development Index (HDI).

```
meta:
  type: alternative
  step: step-combine
  objective: how to combine the three dimensions into a single index?
alternatives:
  - id: arith-mean
    decision: use an arithmetic mean
    rationale: a simple average of the three dimensions
```

3.3 Workflow

The overall workflow for schema-based data analysis is illustrated in Figure 1. A schema can be created either manually from a template or automatically generated by an LLM from a natural-language manuscript. Once created, the schema serves as a structured specification that guides a second LLM in generating the code for the analysis. To improve reliability,

our design include a third LLM agent responsible for code validation. While recent LLMs can generate reasonable code based on prompt, they do not consistently achieve perfect accuracy. The validation process runs the generated script in a sandbox environment and if an error occurs, the third LLM agent is invoked to diagnose the script. The validation agent receives the generated code together with the error message adn teh correponding line number produced during execution, and returns a fixed version of the script. The revised script is then re-executed in the sandbox. This cycle continues until the code runs successfully, a maximum number of attempt is reached, or the user terminates the process. As will be show in Section 4, this validation process resolves the majority of R syntax issues encountered during data wrangling and statistical modelling tasks. On top of supporting a single analytical specification, the workflow also allows users to agument a schema with an alternative YAML file. This expands the original schema into a multiverse of parallel schemas, each representing a different set of decision choices. The same code generation and validation workflow is then applied independently to each schema, producing a corresponding set of validation scripts.

4 Application: many-analysts football study

The many-analysts study posits fixed research questions on a fixed dataset and invites teams of researchers to analyze the problem with whatever software they're comfortable with. In Silberzahn et al. (2018), one question is to answer whether soccer referees are more likely to give red cards to dark-skin-toned player than to light-skin-toned players. The provided dataset contains two raters' rating to the picture of players, the number of redcard received for each player/referee dyade out of the number of games, and a collection of player, referee, and game characteristics: player birthday, height, weight, player position,

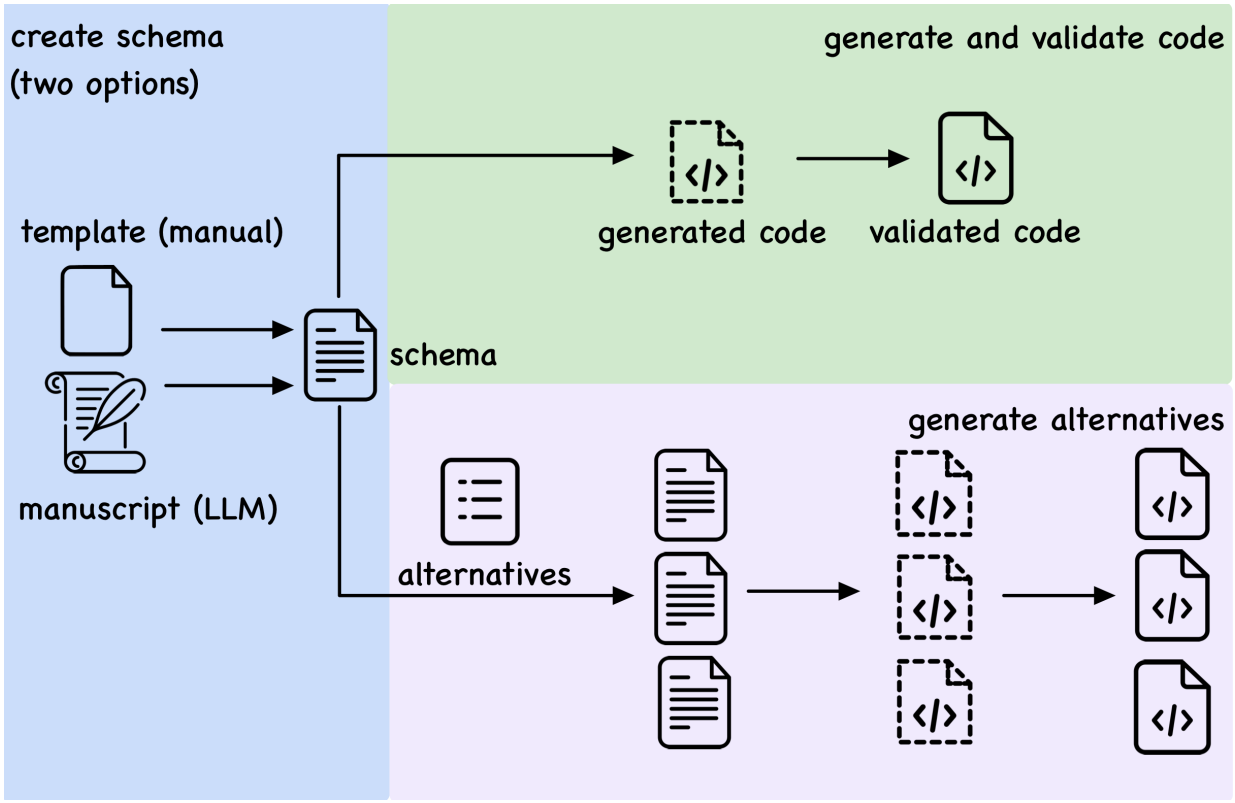


Figure 1: Overview of the schema-driven analysis workflow. Schemas can be authored manually or generated from natural language manuscripts by an LLM. Schemas are used to generate code, which is validated through iterative execution in a sandboxed environment and repaired when errors occur. Alternative specifications can expand a schema into a multiverse of parallel analyses.

league country, number of victories, ties, and defeats, and number of yellow and red cards received. Most groups take a modelling approach to estimate the effect of skin rating on the red card issued with varied distribution assumptions (linear, logistic, and poisson) and covariate choices as fixed and random effects. The study is comprised of an initial analysis conducted separately by each team, followed by a round-robin peer evaluation round, and then each group presents their final analysis after incorporating the feedback from the discussion and presents the results in a way they would in a published article with formal Method section. All the materials related to the study is publicly available on the Open Science Framework (OSF).

From the report presented by each team, we can use LLM to convert the decisions into the schema designed and generate the corresponding coding script. From the write-up reports, we extract the sentences relevant to the decisions made in the final analysis by the team. This step is needed because the original skin rating is 1-5 but later changed to 0-1 and also many teams report their initial analysis and how their final analysis modify on top of the initial analysis. We do this to avoid confusion on reading comprehension with LLM. For general methodology write-up, this step is not necessarily needed [TODO: write it better].

We send the text description of the extractetd decisions for LLM to generate the schema and the code script for each analysis. The code script is run by a separate LLM to validate for potential syntax errors. We manually in-place replace the `glm()` command with `fixest::feglm()` (Bergé et al. 2026) for generalized linear model and the `lme4::glmer()` (Bates et al. 2015) function with `glmmTMB::glmmTMB()` (Brooks et al. 2017) for generalized mix-effect model to speed up the evaluation of eaach script. These two functions are in-place replacements of the classic functions for estimating the relevant model with implementation with multi-threading in C++ (Bergé et al. 2026) and with Template Model Builder

(TMB) (McGillycuddy et al. 2025) for better performance. This replacement should not affect the estimation of the effect size. All the LLM-related simulations are conducted with claude-opus-4.5.

In the validation phase, all team scripts were successfully executed. Notably, 10% of the scripts (n = 2) yielded successful results without requiring any automated syntax corrections by the validator, whereas the vast majority (90%) reached completion after two or three iterations. The most frequent correction involved resolving broken and overflow syntax in comments.

Figure 2 shows the agreement of the estimated effect size, in odd ratio, from our experiments (simulated) and the published effect size in Silberzahn et al. (2018) (published). For most teams, the simulation successfully recovers the odd ratio reported by each analyst team, larger variations are seen in the following teams: Team 1, 9, 15, 21, 23, and 27. Here we listed the reasons for these discrepencies upon investigation:

- **Team 27**: the diffference comes from the ambiguity of the method description, the final approach sections says *In addition to the independent variables specified in the initial approach (skin-tone rating, goals scored, games played, yellow cards, and mean explicit and implicit bias) a variable for player position was included, as well as a variable for the league of the player.* The author is intended to include the interaction effect of rating and position/ league country as other variables in the initial approach, but the LLM interprets it as only include the main effects. This error can be easily caught by comparing the reproducible script provided by the author.
- **Team 9**: the authors mistake player country for referee country in the text description, but their parameter estimation table suggests they are using player country (countryCode - Germany: France) instead.

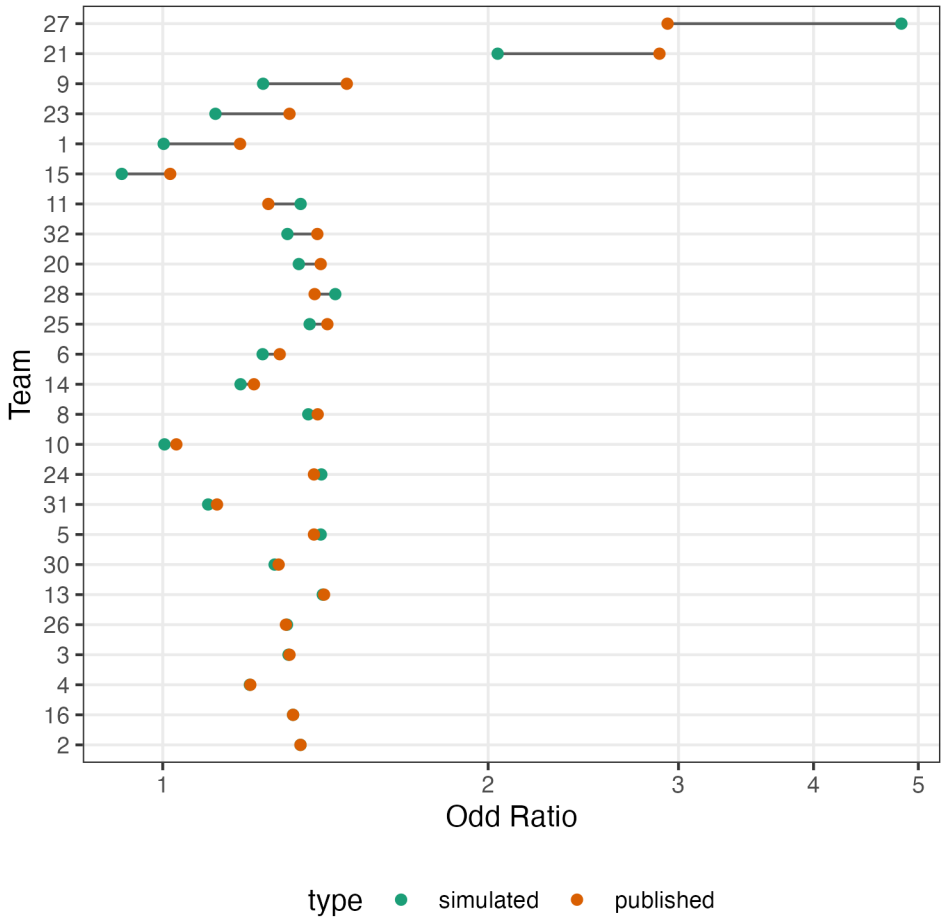


Figure 2: Comparison of the odd ratio of the effect skin color on receiving red cards in football games from our simulation with the schema (simulated) vs. the reported estimated effect size in Silberzahn et al. (2018) (published). Teams are arranged by the difference in odd ratio of our simulation and the reported value. For most teams, the simulation successfully recovers the odd ratio reported by each analyst team, larger variations are seen in the following teams: Team 1, 9, 15, 21, 23, and 27.

- **Team 23 and 1**: the discrepency comes from the LLM fails to properly reshape the data from a player-referee dyad per observation to per game observation. The schema properly identifies this step but the generated code doesn't execute this step properly and create a silent error with the validator.
- **Team 15**: the discrpency comes from the default setting for estimating factor variable in R (our similuation) and SPSS (original script). In SPSS, the factor variables are estimated with effect coding, where the coefficients are compared to the grand mean, rather than to the reference group (dummy coding) in R. After changing the `contrast` options, we obtain the reported odd ratio.

4.1 Exploring alternatives

To demonstrate the functionality for specifying alternatives in Section 3.2, we use Group 5 from the football study above as an example. Group 5 uses a binomial generalized linear mixed model (GLMM) to estimate the effect of skin tone rating on the number of red cards received using a logit link function:

$$\textbf{Model 1 (gm1):} \quad \text{logit}(p_{ijk}) = \beta_0 + \beta_1 X_{ijk} + u_j + v_k$$
$$u_j \sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}})$$

$$\textbf{Model 2 (gm2):} \quad \text{logit}(p_{ijk}) = \beta_0 + \beta_1 X_{ijk} + u_j + v_{0k} + v_{1k} X_{ijk}$$
$$u_j \sim N(0, \sigma^2_{\text{player}}), \quad \mathbf{v}_k \sim N\left(\mathbf{0},\ _{\text{ref}}\right)$$

$$\textbf{Model 3 (gm3):} \quad \text{logit}(p_{ijkm}) = \beta_0 + \beta_1 X_{ijkm} + u_j + v_k + w_{0m} + w_{1m} X_{ijkm}$$
$$u_j \sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}}), \quad \mathbf{w}_m \sim N\left(\mathbf{0},\ _{\text{country}}\right)$$

where p denotes the probability of a player receiving a red card, and X represents the player's skin tone rating. The subscripts i, j, k, and m index the individual observation, player, referee, and the referee's home country, respectively. The parameter β_0 represents the global baseline log-odds of receiving a red card, and β_1 estimates the fixed effect of a player's skin tone. The random structures account for multi-level variation across groupings with varying intercepts and/or slopes: (1) scalar random intercepts (σ^2) capture baseline shifts for players across all specifications ($u_j \sim N(0, \sigma^2_{\text{player}})$) and for individual referees in Models 1 and 3 ($v_k \sim N(0, \sigma^2_{\text{ref}})$); and (2) variance-covariance matrices ( ) accommodate random slopes, allowing the effect of skin tone to vary across individual referees in Model 2 ($\mathbf{v}_k \sim N(\mathbf{0}, \ _{\text{ref}})$) and across referee countries in Model 3 ($\mathbf{w}_m \sim N\left(\mathbf{0}, \ _{\text{country}}\right)$).

The full workflow of experimenting with alternatives is summarized in Figure 3. Alternatives to a specific step are specified in a YAML file in Listing 3, which is then processed to generate multiple copies of the full schema. The code for each schema are then generated separately, and validated to ensure its syntactic correctness and executability. Table 1 summarizes the estimated random effect from the workflow. [TODO: expand more here]

Table 1: Summary of random effect estimates for Group 5. The three models (gm1, gm2, and gm3) represent different specifications of the random effects structure to account for the non-nested cross-classified data structure.

group	term	gm1	gm2	gm3
playerShort	var___(Intercept)	0.3737	0.3738	0.2899
refNum	var___(Intercept)	0.1528	0.1932	0.0748
refNum	var___avgrate01	NA	0.0286	NA
refCountry	var___(Intercept)	NA	NA	0.3990

Table 1: Summary of random effect estimates for Group 5. The three models (gm1, gm2, and gm3) represent different specifications of the random effects structure to account for the non-nested cross-classified data structure.

group	term	gm1	gm2	gm3
refCountry	var___avgrate01	NA	NA	0.0870

5 Discussion

Appendix

5.1 Built-in prompts

Prompt: Extracting the schema from the text

```
You are an expert methodologist and data pipeline architect. I will
provide a text describing a multiverse analysis and a summary of the
actual dataset being used.


=== YOUR TASK ===


Extract a chronological list of methodological decisions (nodes) AND
map the exact data flow (inputs/outputs) for each node
simultaneously.


=== RULES ===
```

Listing 3 Example of a YAML file specifying alternatives for the random effects structure in the many-analyst analysis of Group 5.

```
meta:
  type: alternatives
  step: specify_random_effects_structure
  fork: how to account for the non-nested cross-classified data structure
alternatives:
  - id: "gm1"
    path: "a random effect of player and referee"
    rationale: ""
  - id: "gm2"
    path: "a random effect of player and referee, and a random effect of skin-tone acros
    rationale: ""
  - id: "gm3"
    path: "a random effect of player, referee, and referees' countries of origin, and a
    rationale: ""
```

```
1. THEORY EXTRACTION: For each node, extract:

- 'id': A unique snake_case identifier.

- 'fork': MUST be framed as an open methodological goal that invites
multiple possible approaches. It must NOT describe the final choice.
```



Figure 3: Workflow for generating and validating code for alternatives. The YAML file specifies the alternatives for a specific step, which is then processed to generate multiple copies of the full schema. The code for each schema are then generated separately, and validated to ensure its syntactic correctness and executability.

```
- 'path': A 'path' is strictly a POSITIVE methodological decision
that has potential theoretical alternatives, chosen to resolve the
'fork'.


- 'rationale': WHY that decision (the path) was made, extracted from
the text.


2. DATA MAPPING: Assign 'inputs' (EXACT column names from the dataset
OR outputs from previous nodes) and 'outputs' (invented snake_case
objects like 'df_clean' or 'ranef_spec').
```

3. ANTI-ABSTRACTION (CRITICAL): If the text lists specific variables (e.g., 'centered rater, meanIAT'), DO NOT summarize them away. You MUST capture those specific variables in the 'inputs' array.

4. INLINE ARRAYS: Format arrays strictly on one line: `inputs: [var1, var2]`.

5. CONFIDENCE & CLARIFICATION: Rate your mapping confidence (HIGH, MEDIUM, LOW). If the text abstracts a step and you cannot confidently match it to specific dataset columns, set confidence to LOW and autogenerate a 'clarification_question' asking the user which exact columns to use. If HIGH, output 'null'.

6. Output ONLY valid YAML without markdown formatting.

=== REQUIRED YAML STRUCTURE EXAMPLE ===

meta: type: schema nodes: - fork: variables are in different scales path: apply min-max scaling to each variable justification: to put them on the same scale for combination id: step-scaling confidence: high - fork: combine the school variables into one dimension path: average exp sch and avg sch justification: the most intuitive way id: step-education confidence: low

```
=== DATASET SUMMARY ===


NA


=== METHODOLOGY TEXT ===


[inset text here]
```

Prompt: generating code from the schema

```
You are an expert R programmer. Attached is a text document
containing a SCHEMA that defines a data processing pipeline.


No data file is specified - if data loading is required, use a
sensible placeholder or note it in a comment. Your task is to write a
complete, working R script that implements this pipeline
step-by-step. Use modern R practices (like dplyr or base pipe) and
ensure variables flow correctly from one step to the next as defined
by the inputs and outputs. Output ONLY the complete R script. Do not
start with markdown formatting blocks (like ```R) or backticks.
```

Prompt: validating the code generated

```
You are an expert R developer and automated debugging agent. The user
has provided a data file located at the relative path: `data/[insert
the path to the data]'. Your purpose is to receive broken R scripts,
diagnose the execution error, and return the corrected script.
```

```
CRITICAL INSTRUCTIONS: 1. Fix the bug while preserving the original
intent and logic of the script.  2. If the error is a missing
package, add the necessary `library()` call at the top.  3. If the
error is a missing variable, ensure it is properly initialized before
use.  4. Do NOT hallucinate new data files or external dependencies
unless explicitly provided in the context.

STRICT OUTPUT CONSTRAINTS: - You must output ONLY valid, fully
executable R code.  - Absolutely NO markdown formatting. Do NOT wrap
your response in ```R or ```.  - Absolutely NO conversational filler,
greetings, explanations, or comments about what you fixed.  - The
first character of your response must be R code, and the last
character must be R code.
```

Prompt: generating alternatives for a specific step

```
You are an expert Data Analyst and Methodologist. You are reviewing
an analysis schema to identify "Forking Paths" -- alternative
analytical choices that are equally valid but distinct from the
current approach.

=== DEFINITIONS ===

The schema provided to you consists of steps with:
```

- **ACTION**: The goal of the step (What needs to be done).

- **DECISION**: The specific implementation chosen (How it is done).

- **JUSTIFICATION**: The reasoning behind that decision.

- **ID**: The unique identifier for the step (kebab-case).

=== TASK ===

Focus specifically on the step tagged: "[inset step number here]". Your goal is to generate 3 distinct, valid alternatives for this step.

For each alternative: 1. **Keep the same ACTION** (the goal remains constant).

2. **Change the DECISION** to a different but methodologically sound approach.

3. **Provide a new JUSTIFICATION** explaining why this alternative is valid.

4. **Create a new ID** that reflects the new decision (must be

kebab-case).

=== OUTPUT FORMAT ===

Please output the result in **strictly valid YML format**.

Crucial Formatting Rules:

1. Include a `meta` section at the top with `type: alternative` and the `step`.

2. Output strictly valid YML. All text values (decision, justification) must be enclosed in double quotes ("). Do not use block styles (| or >). Do not wrap lines or insert \n characters within the quotes; keep the text as a single continuous string.

3. Do not include markdown code fences (like ```yml) or conversational text. Just the raw YML.

=== REQUIRED YML STRUCTURE EXAMPLE ===

meta: type: tines_alternative step: [inset step number here] alternatives: - id: step-new-method-name action: Repeat the original action decision: "Description of the new decision..." justification:

```
"This is the reasoning for why this alternative is valid."  - id:
step-another-method ...
```

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